

Quiz 3

BSSE23-A

Multivariable Calculus | (MT101T) | Fall '25

Name: _____ Reg. No.: _____

Q1. [CLO-2] [5 pts]

Sketch the level curve $z = k$ for the specified values of k .

$$z = x^2 + y^2 \text{ for } k = \{-2, -1, 0, 1, 2\}$$

Q2. [CLO-2] [6 pts]

$$\text{Let } f(x, y) = \sin(3x^2 + 6y^2)$$

Find: $f_{xx}, f_{yy}, f_{xy}, f_{yx}$

Q3. [CLO-2] [4 pts]

Show that the ellipsoid: $2x^2 + 3y^2 + z^2 = 9$

and the sphere: $x^2 + y^2 + z^2 - 6x - 8y - 8z + 24 = 0$

have a common tangent plane at the point $(1, 1, 2)$.